

1. Let V be a n -dimensional vector space. Suppose $S = \{v_1, v_2, \dots, v_n\} \subset V$ is a spanning set of V , i.e. $\text{Span}(S) = V$. Prove that S is a basis of V .

Proof: Suppose towards a contradiction that S is not linearly independent. Then there exists a proper subset S_1 of S such that $\text{Span}(S_1) = V$. Since $S_1 \subsetneq S$ is proper, $|S_1| < n$.

Let B be any basis of V . then $|B| = n$.

- As S_1 is a spanning set of V and B is linearly independent, by Replacement Theorem, $|B| \leq |S_1|$.
 2. Consider $V = \mathbb{R}^{n \times n}$ and let $S = \{A \in V : \text{Tr}(A) = 0\}$. $\Rightarrow n = |B| \leq |S_1| < n$. Contradiction

- 1). Prove that S is a subspace of V .

$\Rightarrow S$ is linearly independent and thus a basis for V .

- 2). Find a basis for S . Make sure to justify that the set you give is a basis.

Proof: 1). As $\text{Tr}(0) = 0 \Rightarrow 0 \in S \Rightarrow S \neq \emptyset$.

$$\forall A = (a_{ij}) \text{ and } B = (b_{ij}) \in S \text{ and } \forall c \in \mathbb{R}$$

$$\Rightarrow cA + B = (ca_{ij} + b_{ij})$$

$$\Rightarrow \text{Tr}(cA + B) = \sum_{k=1}^n (ca_{kk} + b_{kk}) = c \left(\sum_{k=1}^n a_{kk} \right) + \left(\sum_{k=1}^n b_{kk} \right)$$

$$= c \text{Tr}(A) + \text{Tr}(B) = c \cdot 0 + 0 = 0$$

$$\Rightarrow cA + B \in S$$

$$\Rightarrow S \text{ is a subspace of } V.$$

- 2). For any $1 \leq k \leq n$, $1 \leq l \leq n$ with $k \neq l$, denote

$$E^{(k,l)} = (e_{ij}) \in \mathbb{R}^{n \times n} \text{ with } e_{ij} = \begin{cases} 1, & (i,j) = (k,l) \\ 0, & (i,j) \neq (k,l) \end{cases}$$

i.e. $E^{(k,l)}$ is the matrix whose (k,l) -th entry is 1, and all other entries are zeros.

For any $1 \leq d \leq n-1$, denote

$$D^{(d)} = (d_{ij}) \text{ with } d_{ij} = \begin{cases} 1, & (i,j) = (d,d) \\ -1, & (i,j) = (n,n) \\ 0, & \text{otherwise.} \end{cases}$$

i.e. $D^{(d)}$ is the matrix whose (d,d) -th entry is 1, (n,n) -th entry is -1, and all other entries are zeros.

Claim: $W = \{ E^{(k,l)} : k, l = 1, \dots, n \text{ with } k \neq l \} \cup \{ D^{(d)} : d = 1, \dots, n-1 \}$
forms a basis of S .

Proof: Define: $T: \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$
 $A \mapsto \text{Tr}(A)$.

Then T is a linear transformation, because for any $A, B \in \mathbb{R}^{n \times n}$
and $c \in \mathbb{R}$: $T(cA+B) = \text{Tr}(cA+B) = c\text{Tr}(A) + \text{Tr}(B) = cT(A) + T(B)$.

② T is also an onto linear transformation, because for any $c \in \mathbb{R}$.

Take $A = cE^{(1,1)}$, then $T(A) = c$.

Then: $S = \ker(T)$.

Then By rank-nullity theorem: $\dim S = \text{nullity}(T) = \dim(\mathbb{R}^{n \times n}) - \dim(\mathbb{R})$
 $= n^2 - 1$.

Note that W has exactly $n^2 - 1$ vectors, thus we only need to
prove that W is linearly independent.

In fact, suppose:

$$\sum_{\substack{k \neq l \\ k=1, \dots, n \\ l=1, \dots, n}} a_{kl} E^{(k,l)} + \sum_{k=1}^{n-1} c_k D^{(k)} = 0$$

$$\Leftrightarrow \begin{pmatrix} c_1 & a_{12} & \dots & a_{1,n-1} & a_{1n} \\ a_{21} & c_2 & \dots & a_{2,n-1} & a_{2n} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n-1,1} & a_{n-1,2} & \dots & c_{n-1} & a_{n-1,n} \\ a_{n1} & a_{n2} & \dots & a_{nn} & (-c_1 - c_2 - \dots - c_{n-1}) \end{pmatrix} = 0 \quad (\text{the zero matrix})$$

$$\Rightarrow c_1 = c_2 = \dots = c_{n-1} = 0$$

and $a_{kl} = 0$ for all $k \neq l$, $k, l = 1, \dots, n$.

$\Rightarrow W$ is linearly independent.

3. Let W_1 and W_2 be subspaces of a vector space V . Let $\dim W_1 = m$ and $\dim W_2 = p$. Define $W_1 + W_2 = \{\mathbf{x}_1 + \mathbf{x}_2 : \mathbf{x}_1 \in W_1, \mathbf{x}_2 \in W_2\}$. Prove that

1). $W_1 + W_2$ is a subspace of V ;

2). $\dim(W_1 + W_2) = m + p - \dim(W_1 \cap W_2)$.

Proof: 1) As $\vec{0} \in W_1$ and $\vec{0} \in W_2 \Rightarrow \vec{0} \in W_1 \cap W_2 \Rightarrow W_1 \cap W_2 \neq \emptyset$.

$\forall \vec{\alpha}, \vec{\beta} \in W_1 + W_2$ and $c \in F$.

$\exists \vec{x}_1, \vec{y}_1 \in W_1$ and $\vec{x}_2, \vec{y}_2 \in W_2$ such that $\vec{\alpha} = \vec{x}_1 + \vec{x}_2$ and $\vec{\beta} = \vec{y}_1 + \vec{y}_2$.

$$\Rightarrow c\vec{\alpha} + \vec{\beta} = c(\vec{x}_1 + \vec{x}_2) + (\vec{y}_1 + \vec{y}_2) = (c\vec{x}_1 + \vec{y}_1) + (c\vec{x}_2 + \vec{y}_2)$$

As $c\vec{x}_1 + \vec{y}_1 \in W_1$ and $c\vec{x}_2 + \vec{y}_2 \in W_2$.

$$(c\vec{x}_1 + \vec{y}_1) + (c\vec{x}_2 + \vec{y}_2) \in W_1 + W_2.$$

$\Rightarrow W_1 + W_2$ is a subspace.

2) Assume $\dim W_1 \cap W_2 = k$. Suppose $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k\}$ is a basis of $W_1 \cap W_2$.

Extend $\{\vec{v}_1, \dots, \vec{v}_k\}$ to a basis of W_1 : $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k, \vec{x}_{k+1}, \dots, \vec{x}_m\}$

Extend $\{\vec{v}_1, \dots, \vec{v}_k\}$ to a basis of W_2 : $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k, \vec{y}_{k+1}, \dots, \vec{y}_p\}$

Claim: $S = \{\vec{v}_1, \dots, \vec{v}_k, \vec{x}_{k+1}, \dots, \vec{x}_m, \vec{y}_{k+1}, \dots, \vec{y}_p\}$ is a basis of $W_1 + W_2$.

Followed by the claim: $\dim W_1 + W_2 = m + p - k = m + p - \dim(W_1 \cap W_2)$.

Proof of claim: ① Need to prove $\text{Span } S = W_1 + W_2$.

$\forall \vec{y} \in W_1 + W_2 \exists \vec{x}_1 \in W_1 = \text{Span}\{\vec{v}_1, \dots, \vec{v}_k, \vec{x}_{k+1}, \dots, \vec{x}_m\}$

and $\vec{x}_2 \in W_2 = \text{Span}\{\vec{v}_1, \dots, \vec{v}_k, \vec{y}_{k+1}, \dots, \vec{y}_p\}$ such that $\vec{y} = \vec{x}_1 + \vec{x}_2$

As: $\exists c_1, \dots, c_m$ and $d_1, \dots, d_p$ such that

$$\vec{x}_1 = c_1 \vec{v}_1 + \dots + c_k \vec{v}_k + c_{k+1} \vec{x}_{k+1} + \dots + c_m \vec{x}_m$$

$$\text{and } \vec{x}_2 = d_1 \vec{v}_1 + \dots + d_k \vec{v}_k + d_{k+1} \vec{y}_{k+1} + \dots + d_p \vec{y}_p$$

$$\Rightarrow \vec{y} = \vec{x}_1 + \vec{x}_2 = (c_1 \vec{v}_1 + \dots + c_k \vec{v}_k + c_{k+1} \vec{x}_{k+1} + \dots + c_m \vec{x}_m) + (d_1 \vec{v}_1 + \dots + d_k \vec{v}_k + d_{k+1} \vec{y}_{k+1} + \dots + d_p \vec{y}_p)$$

$$= (c_1 + d_1) \vec{v}_1 + \dots + (c_k + d_k) \vec{v}_k + c_{k+1} \vec{x}_{k+1} + \dots + c_m \vec{x}_m + d_{k+1} \vec{y}_{k+1} + \dots + d_p \vec{y}_p$$

$$\in \text{Span} \{ \vec{v}_1, \dots, \vec{v}_k, \vec{x}_{k+1}, \dots, \vec{x}_m, \vec{y}_{k+1}, \dots, \vec{y}_p \}$$

$$\Rightarrow W_1 + W_2 \in \text{Span } S.$$

As $S \subseteq W_1 + W_2$ and $W_1 + W_2$ is a subspace (closed under "+" and scalar multiplication)

$$\Rightarrow \text{Span } S \subseteq W_1 + W_2$$

$$\Rightarrow W_1 + W_2 = \text{Span } S.$$

② Need to show S is linearly independent.

$$\text{Suppose } a_1 \vec{v}_1 + \dots + a_k \vec{v}_k + b_{k+1} \vec{x}_{k+1} + \dots + b_m \vec{x}_m + c_{k+1} \vec{y}_{k+1} + \dots + c_p \vec{y}_p = \vec{0}$$

$$\text{Then } a_1 \vec{v}_1 + \dots + a_k \vec{v}_k + b_{k+1} \vec{x}_{k+1} + \dots + b_m \vec{x}_m = -c_{k+1} \vec{y}_{k+1} - \dots - c_p \vec{y}_p$$

$$\text{Denote } \vec{x} = a_1 \vec{v}_1 + \dots + a_k \vec{v}_k + b_{k+1} \vec{x}_{k+1} + \dots + b_m \vec{x}_m = -c_{k+1} \vec{y}_{k+1} - \dots - c_p \vec{y}_p$$

$$\text{Then as } a_1 \vec{v}_1 + \dots + a_k \vec{v}_k + b_{k+1} \vec{x}_{k+1} + \dots + b_m \vec{x}_m \in W_1$$

$$\text{and } \vec{x} = -c_{k+1} \vec{y}_{k+1} - \dots - c_p \vec{y}_p \in W_2$$

$$\Rightarrow \vec{x} \in W_1 \cap W_2$$

$$\Rightarrow \vec{x} = -c_{k+1} \vec{y}_{k+1} - \dots - c_p \vec{y}_p \in W_1 \cap W_2 = \text{Span} \{ \vec{v}_1, \dots, \vec{v}_k \}$$

$$\Rightarrow \exists d_1, \dots, d_k \text{ such that}$$

$$-c_{k+1} \vec{y}_{k+1} - \dots - c_p \vec{y}_p = d_1 \vec{v}_1 + \dots + d_k \vec{v}_k$$

$$\Leftrightarrow d_1 \vec{v}_1 + \dots + d_k \vec{v}_k + c_{k+1} \vec{y}_{k+1} + \dots + c_p \vec{y}_p = \vec{0}$$

Since $\{ \vec{v}_1, \dots, \vec{v}_k, \vec{y}_{k+1}, \dots, \vec{y}_p \}$ is linearly independent

$$d_1 = \dots = d_k = c_{k+1} = \dots = c_p = 0 \quad (**)$$

$$\Rightarrow \vec{x} = a_1 \vec{v}_1 + \dots + a_k \vec{v}_k + b_{k+1} \vec{x}_{k+1} + \dots + b_m \vec{x}_m = -c_{k+1} \vec{y}_{k+1} - \dots - c_p \vec{y}_p = \vec{0}$$

Since $\{\vec{v}_1, \dots, \vec{v}_k, \vec{x}_{k+1}, \dots, \vec{x}_m\}$ is linearly independent

$$a_1 = \dots = a_k = b_{k+1} = \dots = b_m = 0 \quad (**)$$

By (*) & (**), $\{\vec{v}_1, \dots, \vec{v}_k, \vec{x}_{k+1}, \dots, \vec{x}_m, \vec{y}_1, \dots, \vec{y}_p\}$ is linearly independent.

By (1) & (2), $\{\vec{v}_1, \dots, \vec{v}_k, \vec{x}_{k+1}, \dots, \vec{x}_m, \vec{y}_1, \dots, \vec{y}_p\}$ is a basis for $W_1 + W_2$.

4. Consider the following subspaces of $\mathbb{R}^{2 \times 2}$,

$$W_1 = \left\{ \begin{pmatrix} a & b \\ c & a \end{pmatrix} \in \mathbb{R}^{2 \times 2}, a, b, c \in \mathbb{R} \right\}, \quad W_2 = \left\{ \begin{pmatrix} b & a \\ -a & b \end{pmatrix} \in \mathbb{R}^{2 \times 2}, a, b \in \mathbb{R} \right\}.$$

Compute the dimension of the subspace $W_1 + W_2$. Explain your answer. (Note: the definition of $W_1 + W_2$ is given in Problem 3).

Proof: Let $B_1 = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \right\}$. Then B_1 is a basis of W_1 .

In fact, B_1 is linearly independent b/c for any a, b, c such that

$$a \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + c \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix},$$

$$\text{we have} \quad \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} + \begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ c & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\Leftrightarrow \begin{pmatrix} a & b \\ c & a \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \Rightarrow a = b = c = 0$$

$$\textcircled{2} \quad \text{span } B_1 = W_1 \text{ as } \forall \begin{pmatrix} a & b \\ c & a \end{pmatrix} = a \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + c \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

$$\Rightarrow \dim W_1 = 3.$$

Let $B_2 = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right\}$. Then B_2 is a basis of W_2 .

In fact, B_2 is linearly independent b/c for any a, b such that

$$a \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} a & b \\ -b & a \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \Rightarrow a = b = 0$$

$$\text{span } B_2 = W_2 \text{ b/c } \forall \begin{pmatrix} b & a \\ -a & b \end{pmatrix} = b \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + a \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$\Rightarrow \dim W_2 = 2.$$

Notice that $W_2 \subseteq W_1$ thus $W_1 \cap W_2 = W_2$

$$\Rightarrow \dim W_2 = 2 \quad 3 \text{ of } 7$$

By the result from #3: $\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2)$
 $= 3 + 2 - 2 = 3.$

5. Show that the polynomials $2, 1+t, t+t^2$ form a basis for $P^2(\mathbb{R})$. Then find the coordinate of $3+t+2t^2$ in this basis.

Proof. ① Suppose $a(2) + b(1+t) + c(t+t^2) = 0$

$$\Leftrightarrow (2a+b) + (b+c)t + c(t^2) = 0$$

$$\Leftrightarrow \begin{array}{l} 2a+b = 0 \\ b+c = 0 \\ c = 0 \end{array} \Leftrightarrow a=b=c=0$$

$\Rightarrow \{2, 1+t, t+t^2\}$ is linearly independent

Since $\dim P^2(\mathbb{R}) = 3$. and there are exactly 3 vectors in $\{2, 1+t, t+t^2\}$

$\Rightarrow \{2, 1+t, t+t^2\}$ is a basis of $P(\mathbb{R})$

② Suppose $3+t+2t^2 = a(2) + b(1+t) + c(t+t^2)$

$$3+t+2t^2 = (2a+b) + (b+c)t + ct^2$$

$$\Rightarrow \begin{array}{l} 2a+b = 3 \\ b+c = 1 \\ c = 2 \end{array} \Rightarrow \begin{array}{l} a = 2. \\ b = -1 \\ c = 2 \end{array}$$

$\Rightarrow$ the coordinate of $3+t+2t^2$ under the basis $\{2, 1+t, t+t^2\}$

is $\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$.

6. Let $V = \mathbb{R}^\infty = \{(a_1, a_2, \dots) : a_1, a_2, \dots \in \mathbb{R}\}$. Define $T : V \rightarrow V$ by

$$T((a_1, a_2, a_3, \dots)) = (a_2, a_3, \dots).$$

1). Prove that T is a linear transformation on V .

2). Prove that T is onto, but not one-to-one.

1). $\forall (a_1, a_2, a_3, \dots), (b_1, b_2, b_3, \dots) \in V$ and $\forall c \in \mathbb{R}$

$$c(a_1, a_2, a_3, \dots) + (b_1, b_2, b_3, \dots) = (ca_1 + b_1, ca_2 + b_2, ca_3 + b_3, \dots)$$

$$\text{Then } T(c(a_1, a_2, a_3, \dots) + (b_1, b_2, b_3, \dots)) = (ca_2 + b_2, ca_3 + b_3, \dots)$$

$$= c(a_2, a_3, \dots) + (b_2, b_3, \dots) = cT((a_1, a_2, \dots)) + T(b_1, b_2, \dots)$$

$\Rightarrow T$ is a linear transformation.

2). $\forall (b_1, b_2, \dots) \in \mathbb{R}^\infty$, take $\vec{x} = (1, b_1, b_2, \dots) \in \mathbb{R}^\infty$ Then $T((1, b_1, b_2, \dots)) = (b_1, b_2, \dots)$

$\Rightarrow T$ is onto.

Take $\vec{x}_1 = (1, a_1, a_2, a_3, \dots)$ and $\vec{x}_2 = (0, a_1, a_2, a_3, \dots)$, then $\vec{x}_1 \neq \vec{x}_2$, but $T(\vec{x}_1) = T(\vec{x}_2) = (a_1, a_2, \dots)$

7. Let $V = \mathbb{R}^\infty = \{(a_1, a_2, \dots) : a_1, a_2, \dots \in \mathbb{R}\}$. Define $T : V \rightarrow V$ by

$$T((a_1, a_2, a_3, \dots)) = (0, a_1, a_2, \dots).$$

$\Rightarrow T$ is not one-to-one

1). Prove that T is a linear transformation on V .

2). Prove that T is one-to-one, but not onto.

1). $\forall (a_1, a_2, a_3, \dots)$ and $(b_1, b_2, b_3, \dots) \in \mathbb{R}^\infty$ and $c \in \mathbb{R}$.

$$T(c(a_1, a_2, a_3, \dots) + (b_1, b_2, b_3, \dots)) = T((ca_1 + b_1, ca_2 + b_2, ca_3 + b_3, \dots)) = (0, ca_1 + b_1, ca_2 + b_2, \dots)$$

$$= c(0, a_1, a_2, \dots) + (0, b_1, b_2, \dots) = cT((a_1, a_2, a_3, \dots)) + T((b_1, b_2, b_3, \dots))$$

$\Rightarrow T$ is a linear transformation.

2). Let $\vec{x} = (a_1, a_2, a_3, \dots)$ and $\vec{y} = (b_1, b_2, b_3, \dots)$ such that $T(\vec{x}) = T(\vec{y})$

$$\Rightarrow T(\vec{x}) = (0, a_1, a_2, a_3, \dots) = T(\vec{y}) = (0, b_1, b_2, b_3, \dots)$$

$$\Rightarrow a_1 = b_1, a_2 = b_2, a_3 = b_3, \dots$$

$$\Rightarrow \vec{x} = \vec{y}$$

$\Rightarrow T$ is one-to-one.

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T is not onto because $(1, 0, 0, \dots) \in \mathbb{R}^\infty$ is not an image of any vector in $\mathbb{R}^\infty$ under T .

8. Let V and W be vector spaces over F . Let $\mathcal{L}(V, W)$ be the set of all linear transformations from V to W . For any $T, U \in \mathcal{L}(V, W)$, define $T + U$ by

$$(T + U)(\mathbf{x}) = T(\mathbf{x}) + U(\mathbf{x}), \text{ for any } \mathbf{x} \in V.$$

For any $T \in \mathcal{L}(V, W)$ and $c \in F$, define cT by

$$(cT)(\mathbf{x}) = cT(\mathbf{x}), \text{ for any } \mathbf{x} \in V.$$

Prove that $\mathcal{L}(V, W)$ with the above addition and scalar multiplication is a vector space over F .

Proof: 0): $\forall U, T \in \mathcal{L}(V, W)$ and any $\vec{x}, \vec{y} \in V$, and any $c \in \mathbb{R}$ (or $\mathbb{C}$)

$$\begin{aligned} (U+T)(c\vec{x} + \vec{y}) &= U(c\vec{x} + \vec{y}) + T(c\vec{x} + \vec{y}) \\ &= cU(\vec{x}) + U(\vec{y}) + cT(\vec{x}) + T(\vec{y}) \\ &= [cU(\vec{x}) + cT(\vec{x})] + [U(\vec{y}) + T(\vec{y})] \\ &= [c(U+T)(\vec{x})] + [(U+T)(\vec{y})] \end{aligned}$$

$\mathcal{L}(V, W)$ is closed under "+"

$\Rightarrow U+T$ is also a linear transformation

$\Rightarrow U+T \in \mathcal{L}(V, W)$.

$\mathcal{L}(V, W)$ is closed under scalar multiplication

$\forall T \in \mathcal{L}(V, W)$ and $\forall \lambda \in \mathbb{R}$ (or $\mathbb{C}$), and $\forall \vec{x}, \vec{y} \in V$ and $c \in \mathbb{R}$ (or $\mathbb{C}$)

$$\begin{aligned} (\lambda T)(c\vec{x} + \vec{y}) &= \lambda (T(c\vec{x}) + T(\vec{y})) \\ &= \lambda T(c\vec{x}) + \lambda T(\vec{y}) \\ &= c \cdot (\lambda T)(\vec{x}) + (\lambda T)(\vec{y}) \end{aligned}$$

$\Rightarrow \lambda T$ is a linear transformation

$\Rightarrow \lambda T \in \mathcal{L}(V, W)$

1): $\forall \vec{x} \in V: (U+T)(\vec{x}) = U(\vec{x}) + T(\vec{x}) = T(\vec{x}) + U(\vec{x}) = (T+U)(\vec{x})$

$\Rightarrow U+T = T+U$

I). $\forall U, T, S \in \mathcal{L}(V, W): \forall \vec{x} \in V:$

$$[(U+T)+S](\vec{x}) = U(\vec{x}) + T(\vec{x}) + S(\vec{x}) = (U+(S+T))(\vec{x})$$

$$\Rightarrow (U+T)+S = U+(T+S).$$

II). Denote the zero linear transformation by: $T_0: V \rightarrow W$
 $\vec{v} \mapsto \vec{0}$

Then $\forall T \in \mathcal{L}(V, W)$ and $\forall \vec{x} \in V:$

$$(T+T_0)(\vec{x}) = T(\vec{x}) + T_0(\vec{x}) = T(\vec{x}) + \vec{0} = T(\vec{x}) = T_0(\vec{x}) + T(\vec{x}) = (T_0+T)(\vec{x})$$

$$\Rightarrow T+T_0 = T = T_0+T.$$

IV) $\forall T \in \mathcal{L}(V, W)$ and $\forall \vec{x} \in V:$

define $T': V \rightarrow W$
 $\vec{x} \mapsto -T(\vec{x}).$

then $T' \in \mathcal{L}(V, W)$
(one may need to verify this assertion).

$$\Rightarrow (T+T')(\vec{x}) = T(\vec{x}) + T'(\vec{x}) = T(\vec{x}) - T(\vec{x}) = \vec{0} = T_0(\vec{x}).$$

$$\Rightarrow T+T' = T_0.$$

V). $\forall T \in \mathcal{L}(V, W):$

$$\forall \vec{x} \in V: (1 \cdot T)(\vec{x}) = 1 \cdot T(\vec{x}) = T(\vec{x}) \Rightarrow 1 \cdot T = T.$$

VI): $\forall a, b \in \mathbb{R}$ (or $\mathbb{C}$) and $T \in \mathcal{L}(V, W)$

$$[(ab)T](\vec{x}) = (ab)T(\vec{x}) = a(bT(\vec{x})) = a(bT)(\vec{x})$$

$$\Rightarrow (ab)T = a(bT)$$

VII) $\forall a, b \in \mathbb{R}$ (or $\mathbb{C}$) and $\forall T \in \mathcal{L}(X, Y)$

$$\begin{aligned} \forall \vec{x} \in V: [(a+b)T](\vec{x}) &= (a+b)T(\vec{x}) = aT(\vec{x}) + bT(\vec{x}) \\ &= (aT + bT)(\vec{x}) \end{aligned}$$

$$\Rightarrow (a+b)T = aT + bT.$$

VIII) $\forall a \in \mathbb{R}$ (or $\mathbb{C}$), and $\forall U, T \in \mathcal{L}(V, W)$

$$\begin{aligned} \forall \vec{x} \in V: [a(U+T)](\vec{x}) &= a[(U+T)(\vec{x})] = a(U(\vec{x}) + T(\vec{x})) = aU(\vec{x}) + aT(\vec{x}) \\ &= (aU + aT)(\vec{x}) \end{aligned}$$

$$\Rightarrow a(U+T) = aU + aT.$$

$\Rightarrow \mathcal{L}(V, W)$ is a vector space.

9. True or false. (No explanation needed)

- F** 1). If S is a linear dependent set, then each vector in S is a linear combination of other vectors in S .
- T** 2). Any set containing the zero vector is a linearly dependent.
- T** 3). Subset of linearly independent set is linearly independent. (Note, empty set is linearly independent)
- F** 4). Let V be a vector space. Let $W \subseteq V$ be a subspace with $\dim W = \dim V$. Then $W = V$.